FinDoc-AuditEngine v3.0 — Prototype Architecture & Technical Specification

Project Classification: Internship Capstone Working Prototype / Production-Oriented Reference Implementation

Currency Display: USD ($) and INR (`1 USD = 86.50 INR` configured conversion rate)

Verified Automated Tests: Day 29: `28 passed` (`python -m pytest tests -q`) | Day 30: `33 passed` (`python -m pytest tests -q`) — *all collected tests passed*

1. Architectural Overview & Prototype Scope

`FinDoc-AuditEngine v3.0` is a modular FastAPI prototype designed to demonstrate how deterministic procurement accounting rules can be combined with pre-inference regex privacy masking and unsupervised anomaly scoring.

```
flowchart LR
    A["Client / Web Workbench<br/>(Structured JSON or<br/>Raw Invoice Text)"] --> B["Stage 1: Regex PII Masking
<br/>(FinancialPrivacyScrubber)<br/>6 Configured Patterns"]
    B --> C["Stage 2: 3-Way Matcher<br/>(ReconciliationMatcher)<br/>Invoice vs. PO vs. GRN"]
    C --> D["Stage 3: Anomaly Scorer<br/>(AnomalyScorer)<br/>7-D Synthetic Baseline IF<br/>+ Perturbation Attri
bution"]
    D --> E["Stage 4: Governance Router<br/>(ApprovalRouter)<br/>5 Approval Tiers"]
    E --> F["Prototype Storage<br/>(In-Memory AUDIT_LEDGER<br/>Python List + CSV Export)"]
```

2. Detailed Stage Breakdown

Stage 1: Regex-Based Pre-Inference Privacy Masking (`app/core/privacy.py`)

- `FinancialPrivacyScrubber.scrub_text()`: Scans input strings against **6 configured regular expression patterns** before logging or downstream processing:

1. `EMAIL` (`[A-Za-z0-9._%+-]+@[A-Za-z0-9.-]+\.[A-Za-z]{2,7}`)
2. `PAN` (Indian Permanent Account Number format: `[A-Z]{5}[0-9]{4}[A-Z]{1}`)
3. `CREDIT_CARD` (16-digit card sequences)
4. `ACCOUNT` (8–18 digit numbers preceded by `account`, `acct`, `iban`, `acc`, or `bank`)
5. `PHONE` (10-digit Indian mobile numbers starting with `6–9`, optional `+91`)
6. `AADHAAR` (12-digit format `[1-9]\d{3}\s?\d{4}\s?\d{4}`)

- **Invoice-Text Parser (`POST /api/v1/audit/parse-raw`):** Applies regex extraction on raw text memos to populate Invoice ID, PO Number, Vendor Name, Country code, and USD total while masking matched PII tokens into indexed surrogates (`[EMAIL_01]`, `[PAN_02]`). *Note: This endpoint performs regex-based invoice-text parsing on character strings; it does not integrate an optical character recognition (OCR) engine for scanned images/PDFs.*

Stage 2: Deterministic 3-Way Reconciliation (`app/services/matcher.py`)

- `ReconciliationMatcher.match()`: Enforces the configured SKU-level pricing, dock quantity, and subtotal rules between an `InvoicePayload` and a `PurchaseOrderRecord`:
  - **Unit Rate Check:** Flags `RATE_VARIANCE` if `invoice_item.unit_price - po_item.unit_price > $0.01`.
  - **Dock GRN Net Quantity Check:** Computes net accepted units $Q_{\text{net}} = Q_{\text{received}} - Q_{\text{rejected}}$ and flags `QTY_SHORTAGE` if billed quantity exceeds Q_{net}.
  - **Subtotal Tolerance:** Enforces `SUBTOTAL_TOLERANCE = 0.05` (`$0.05`) to prevent binary floating-point rounding discrepancies on tax calculations from causing false rejections.

Stage 3: 7-Feature Unsupervised Anomaly Scoring & Perturbation Attribution (`app/services/anomaly_detector.py`)

> ***Important Model Scope Statement:*** *The Isolation Forest model (`n_estimators=200`, `StandardScaler`, `random_state=42`) is trained on a reproducible seeded synthetic baseline of 1,000 transaction-like records to demonstrate financial anomaly scoring. Its thresholds and risk behavior would require evaluation and calibration against representative, authorized real-world data before production use. Furthermore, the displayed `0–100` risk score is an engineered anomaly-risk indicator derived from tree isolation depth, not a calibrated probability of fraud.*

Index	Feature Key	Mathematical Definition	Synthetic Baseline Role		
0	`log_total_usd`	**$\ln(1 + \text{Total}_{USD})$**	Compresses heavy-tailed invoice dollar totals so magnitude does not dominate bounded ratios.		
1	`price_variance_ratio`	$\Delta P_{\text{USD}} / \max(\text{Subtotal}_{\text{PO}}, 1)$	Captures relative unit-price markup above PO contract rates.		
2	`qty_shortage_ratio`	$Q_{\text{short}} / \max(Q_{\text{billed}}, 1)$	Captures fraction of billed units missing or rejected at the receiving dock.		
3	`tax_rate_deviation`	$	\tau_{\text{inv}} - 0.18	$	Captures deviation from the standard 18% GST benchmark.

Index	Feature Key	Mathematical Definition	Synthetic Baseline Role		
4	`line_item_concentration`	$\max_i(\text{LineTotal}_i) / \text{Subtotal}_{\text{inv}}$	Distinguishes single-SKU lump-sum charges from multi-line orders.		
5	`jurisdiction_risk_weight`	`0.05` (Domestic/Standard) to `1.00` (BZ, KY, VG, PA)	Encodes configured vendor country risk tier.		
6	`payment_urgency_index`	$\min(1.0, 15.0 / \max(\text{payment_days}, 1))$	Flags rush wire demands (`Net-1` to `Net-3`) vs. standard `Net-30`.		

- **Local Perturbation Attribution:** For each feature $k \in \{0,\dots,6\}$, the scorer replaces x_k with its synthetic baseline median **mu_k** and computes the marginal score difference **Delta R_k = max(0, R(x) - R($x^{(k <- mu_k)}$))** to surface the top contributing features in the UI.

Stage 4: Governance Routing & In-Memory Prototype Ledger (`app/services/router.py` & `app/routers/audit.py`)

- `ApprovalRouter.route_decision()`: Assigns one of 5 governance tiers (`TIER_1_STP_AUTO_APPROVED`, `TIER_2_MANAGER_REVIEW`, `TIER_3_DIRECTOR_SIGNOFF`, `REJECTED_DISCREPANCY`, or `FROZEN_FRAUD_RISK`).
- **In-Memory Prototype Audit Ledger (`AUDIT_LEDGER`):** Decisions are stored in an in-memory Python list during runtime for dashboard filtering and CSV export (`GET /api/v1/audit/export/csv`).

3. Current Prototype Limitations & Planned Future Work

Area	Current Prototype Implementation	Planned Production Improvement (Future Work)
Audit Ledger Storage	Held in an in-memory Python list (`AUDIT_LEDGER`); records are lost when the server process restarts.	Migrate to durable **PostgreSQL (`SQLAlchemy 2.0` + `asyncpg`)** tables with **Redis** idempotency keys.
ML Training Data & Calibration	Fit on a seeded synthetic baseline of 1,000 records; outputs an uncalibrated `0–100` anomaly indicator.	Train and validate on authorized historical enterprise AP data; calibrate probabilities and monitor drift (**EvidentlyAI PSI**).
Document Ingestion	Accepts structured JSON or performs regex invoice-text parsing (`/api/v1/audit/parse-raw`).	Integrate optical OCR and layout-aware document extraction (**LayoutLMv3** / PDF table parsing).
PII Masking Coverage	Uses 6 deterministic regex patterns; may miss non-standard formats or free-form names/addresses.	Combine regex rules with context-aware Named Entity Recognition (**Presidio / Transformer NER**).
Security & Access Control	Local prototype endpoints without user authentication or role-based authorization.	Add **OAuth2 / JWT** authentication, role-based access control (RBAC), and TLS termination.